

19 — Step-by-Step Live Project Demonstration Script

This document provides a live demonstration script for presenting the Maritime Port Intelligence Platform (MPIP) to an audience or evaluation panel.

■ Pre-Demo Setup Verification Checklist

1. **Backend Server:** Ensure FastAPI backend is running on **CODE_PH_0**.

```
python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8000
```

2. **Frontend Server:** Ensure Streamlit frontend is running on **CODE_PH_0**.

```
streamlit run frontend/app.py
```

3. **Browser Tab:** Open **CODE_PH_0** in full-screen browser view.

■ Live Demo Step-by-Step Guide

STEP 1 — Platform Overview & Core Objective Header

- **SHOW ON SCREEN:**

- The top header card (**CODE_PH_0**).
- The top hero positioning banner (**CODE_PH_0**).

- **WHAT TO SAY:**

*"Welcome everyone. As you can see on the screen, our platform's primary objective is **Predictive Cargo Analytics**. Rather than simply tracking ship coordinates, our platform connects historical cargo data, machine learning algorithms, commodity demand trends, and anomaly detection to support proactive port planning decisions."*

STEP 2 — Core Feature: ■ Cargo Forecasting Tab

- **SHOW ON SCREEN:**

- Click on the 3rd navigation tab: **CODE_PH_0**.
- Point out the 3 visual capability cards: **CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**.

- **WHAT TO SAY:**

*"Here on the Cargo Forecasting Command Center, we immediately see our three core operational pillars. Let's select a specific commodity to analyze, such as **TOTAL COAL**."*

STEP 3 — Commodity Selection & Metric Snapshot

- **SHOW ON SCREEN:**

- Select **CODE_PH_0** from the commodity dropdown.
- Point to the **Selected Commodity Analytics Banner** and **NMPA Facility Banner**.

- **WHAT TO SAY:**

"When I select 'TOTAL COAL', the platform dynamically pulls multi-year historical records for New Mangalore Port Authority. We see our current monthly volume is 625,768 tonnes, with an expected 6-month average of 710,889 tonnes—a +13.6% growth trend. Notice the NMPA facility mapping below: thermal coal is automatically mapped to Mechanized Coal Handling Berths 15 & 16 (14.0m draft depth) supplying the UPCL Power Plant."

STEP 4 — Main Forecast Chart & 95% Confidence Interval

- **SHOW ON SCREEN:**

- Hover over the Plotly main time-series forecast graph (**CODE_PH_0**).
- Highlight the dark grey historical line, the dashed gold forecast line, and the shaded 95% dynamic prediction interval band.

- **WHAT TO SAY:**

"On this chart, the dark grey line represents historical actual cargo tonnage. The dashed gold line displays our machine learning forecast for the next 6 months. The shaded amber band represents our 95% dynamic prediction interval, giving port operators mathematical confidence bounds for berth capacity planning."

STEP 5 — Explainable Forecast Drivers

- **SHOW ON SCREEN:**

- Scroll down to **CODE_PH_0**.

- **WHAT TO SAY:**

*"Crucially, our platform is not a black box. Here we answer: **WHY is cargo volume expected to change?** The ML driver engine decomposes the forecast into weighted factors—showing that UPCL power plant demand contributes 45% to the forecast, while seasonal monsoon swell delays account for 25%."*

STEP 6 — Model Validation & Chronological Backtesting

- **SHOW ON SCREEN:**

- Scroll down to **CODE_PH_0**.
- Highlight the WAPE metric (**CODE_PH_0**) and model accuracy (**CODE_PH_1**).

- **WHAT TO SAY:**

"To prove the mathematical accuracy of our model, we run out-of-sample chronological backtesting against actual test data. We compare our Primary ML Model against a Seasonal Naive Baseline Model. Our model achieves an 82.8% accuracy score evaluated using Weighted Absolute Percentage Error (WAPE)."

STEP 7 — Interactive What-If Cargo Scenario Simulator

- **SHOW ON SCREEN:**

- Move the **Vessel Arrival Change** slider to **CODE_PH_0**.
- Move the **Trade Demand Shift** slider to **CODE_PH_0**.
- Click **Run Cargo Scenario Simulation**.
- Point to the updated metric: **CODE_PH_0**.

- **WHAT TO SAY:**

"Now let's test a contingency scenario. Suppose vessel fleet arrivals surge by 15% and trade demand increases by 10%. When I click 'Run Cargo Scenario Simulation', the backend recalculates throughput in real time, projecting a net volume increase of 571,555 tonnes and issuing an operational advisory to deploy two additional bulk unloading cranes."

STEP 8 — Supporting Modules: Trade, Anomaly & Copilot

- **SHOW ON SCREEN:**

- Briefly click **Trade Intelligence** (**CODE_PH_0**).
- Briefly click **Anomaly Detection** (**CODE_PH_0**).
- Briefly click **AI Maritime Copilot** and submit a sample query: "Why is cargo forecast increasing?".

- **WHAT TO SAY:**

"Supporting tabs reinforce our cargo focus. Trade Intelligence tracks commodity demand across major trade lanes. Anomaly Detection classifies cargo volume surges and decline risks with a 4-tier severity scale. Finally, our AI Maritime Copilot uses a 3-tier cognitive dispatcher to answer natural language query requests instantly."

STEP 9 — Concluding Remarks

- **WHAT TO SAY:**

"In conclusion, our platform transforms fragmented maritime data into actionable, predictive decision support. Thank you, and I am now ready to take your questions."